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NEGATIVE ELECTRODE PLATE, SODIUM-ION BATTERY, AND ELECTRONIC DEVICE

Abstract

Provided are a negative electrode plate, a sodium-ion battery, and an electronic device. The negative electrode plate includes a negative electrode current collector and a carbon coating disposed on at least a portion of the surface of the negative electrode current collector. The carbon coating includes a carbon material, a solid electrolyte, and a polymer binder. A surface of the carbon coating facing away from the negative electrode current collector has holes extending along the thickness direction of the carbon coating. A depth of the holes is smaller than a thickness of the carbon coating.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202410268840.X, filed on Mar. 8, 2024, which is incorporated herein by reference in its entirety.

FIELD

[0002] The present disclosure relates to the technical field of sodium-ion battery, and specifically, to a negative electrode plate, a sodium-ion battery, and an electronic device.

BACKGROUND

[0003] With the expanding use of lithium-ion batteries in the energy storage market, the industry is rapidly transitioning into the terawatt-hour (TWh) era, where supply chain security has become a critical concern of the TWh era, especially the emerging challenge of limited availability of lithium resources. Sodium-ion battery, known for the abundant raw material supply and low-cost advantages thereof, has garnered increasing attention. However, since metallic sodium has higher reduction potential and greater molecular weight than metallic lithium, the energy density of sodium-ion batteries with similar working principles is significantly lower than that of lithium-ion batteries. In addition, due to the greater ionic radius of sodium ions, a greater volumetric expansion may occur during intercalation and deintercalation of sodium ions, leading to reduced cycle reversibility, low energy density, and rapid capacity degradation, which restrict the practical application thereof.

[0004] To improve energy density of the cell, the sodium-ion battery without a negative active material has emerged as an ultimate goal for enhancing the energy density. However, the battery without a negative active material has relatively poor capabilities of liquid absorption and retention at the negative electrode side, resulting in an acceleration of the decay of the cell during cycling.

SUMMARY

[0005] In a first aspect of the present disclosure, the present disclosure provides a negative electrode plate. The negative electrode plate includes a negative electrode current collector and a carbon coating disposed on at least a portion of a surface of the negative electrode current collector. The carbon coating includes a carbon material, a solid electrolyte, and a polymer binder. A surface of the carbon coating facing away from the negative electrode current collector has holes extending along the thickness direction of the carbon coating. A depth of the holes is smaller than a thickness of the carbon coating.

[0006] In a second aspect of the present disclosure, the present disclosure provides a sodium-ion battery. The sodium-ion battery includes a positive electrode plate, a separator, a negative electrode plate, and an electrolyte solution. The negative electrode plate is the negative electrode plate according to the first aspect of the present disclosure.

[0007] In a third aspect of the present disclosure, the present disclosure provides an electric device. The electric device includes the sodium-ion battery according to the second aspect of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The aforementioned and/or additional aspects and advantages of the present disclosure will become more apparent and readily understood from the following detailed description in conjunction with the accompanying figures, in which:

[0009] FIGURE is a schematic structural diagram of a negative electrode plate according to an embodiment of the present disclosure.

REFERENCE CHARACTER

[0010] **1**—Negative electrode current collector; **2**—Carbon coating; **3**—Holes.

DETAILED DESCRIPTION

[0011] Embodiments of the present disclosure are described in detail below. An example of the embodiments is illustrated in the accompanying drawings, in which the same or similar reference character throughout denotes the same or similar elements or elements having the same or similar functions. The embodiments described below by reference to the accompanying drawings are illustrative and are intended to explain, rather than limiting, the present disclosure.

[0012] The present disclosure is intended to address at least one of the technical problems in the related art to some extent. Therefore, the present disclosure provides a negative electrode plate, a sodium-ion battery, and an electronic device. In the present disclosure, by providing holes in a surface of the negative electrode plate, a contact angle between an electrolyte solution and a negative electrode plate surface can be effectively reduced, while increasing a surface roughness of the negative electrode plate, enabling a sodium metal layer deposited on a surface of the negative electrode plate to have a stronger bonding force to the surface of the negative electrode plate. Furthermore, an electric field on the negative electrode plate surface can be adjusted, and the uniformity and compactness of the deposited sodium on the negative electrode plate surface can be improved. Additionally, the carbon coating contains a solid electrolyte, which can further enhance an affinity between the carbon coating and the electrolyte solution, thereby improving a wettability and ionic migration rate on the negative electrode side.

[0013] In an aspect of the present disclosure, the present disclosure provides a negative electrode plate. According to an embodiment of the present disclosure, the negative electrode plate includes a negative electrode current collector and a carbon coating disposed on at least a portion of a surface of the negative electrode current collector. The carbon coating includes a carbon material, a solid electrolyte, and a polymer binder. A surface of the carbon coating facing away from the negative electrode current collector has holes extending along the thickness direction of the carbon coating. A depth of the holes is smaller than a thickness of the carbon coating.

[0014] For the negative electrode plate according to an embodiment of the present disclosure, by providing the holes in the surface of the negative electrode plate, a contact angle between an electrolyte solution and a negative electrode plate surface can be effectively reduced, which enables the electrolyte solution to rapidly wet the surface of the negative electrode plate, thereby significantly enhancing the liquid absorption and retention capability of the negative electrode plate and reducing the ionic transport impedance on the negative electrode side. Meanwhile, by providing the holes in the surface of the negative electrode plate, a surface roughness of the negative electrode plate is increased, enabling a sodium metal layer deposited on a surface of the negative electrode plate to have a stronger bonding force to the surface of the negative electrode plate. In this way, issues such as peeling and pulverization of the sodium deposition layer during cycling can be prevented, and the irreversible capacity loss can be avoided. Furthermore, an electric field on the negative electrode plate surface can be adjusted, and the uniformity and compactness of the deposited sodium on the negative electrode plate surface can be improved. Additionally, the solid electrolyte in the carbon coating can further enhance an affinity between the carbon coating and the electrolyte solution, thereby improving a wettability and ionic migration rate on the negative electrode side.

[0015] Furthermore, the negative electrode plate of the above embodiments of the present disclosure may further have the following additional technical features.

[0016] In some embodiments of the present disclosure, the depth of the holes is denoted as $h_{sub.2}$, and the thickness of the carbon coating is denoted as $h_{sub.1}$, where $h_{sub.2}$ and $h_{sub.1}$ satisfy: $40\% \leq h_{sub.2}/h_{sub.1} \leq 80\%$.

[0017] In some embodiments of the present disclosure, the holes have each a diameter ranging from 1 μm to 3 μm ; and/or a density of the holes on the surface of the carbon coating ranges from 1,000 holes/ mm^2 to 3,000 holes/ mm^2 .

[0018] In some embodiments of the present disclosure, a contact angle θ between a surface of the negative electrode plate and an electrolyte solution satisfies: $0^\circ \leq \theta \leq 10^\circ$.

[0019] In some embodiments of the present disclosure, a roughness R_a of a surface of the negative electrode plate satisfies: $0.4 \mu\text{m} \leq R_a \leq 3.2 \mu\text{m}$.

[0020] In some embodiments of the present disclosure, the thickness h_1 of the carbon coating is not greater than 5 μm .

[0021] In some embodiments of the present disclosure, a weight ratio of the carbon material, the solid electrolyte, and the polymer binder is (60 to 80):(0.5 to 10):(10 to 30).

[0022] In some embodiments of the present disclosure, the carbon material is a combination of a one-dimensional carbon material and at least one selected from zero-dimensional carbon material, two-dimensional carbon material, and three-dimensional carbon material.

[0023] In some embodiments of the present disclosure, the solid electrolyte includes at least one of $\beta''\text{-Al}_{sub.2}\text{O}_{sub.3}$, $\text{Na}_{sub.3}\text{Zr}_{sub.2}\text{Si}_{sub.2}\text{PO}_{sub.12}$, $\text{Na}_{sub.3+x}\text{La}_{sub.x}\text{Zr}_{sub.2x}\text{Si}_{sub.2}\text{PO}_{sub.12}$, $\text{Na}_{sub.3}\text{PS}_{sub.4}$, $\text{Na}_{sub.11}\text{Sn}_{sub.2}\text{PS}_{sub.4}$, or $\text{Na}_{sub.7}\text{P}_{sub.3}\text{S}_{sub.11}$, where $0 < x \leq 0.5$.

[0024] In some embodiments of the present disclosure, the carbon material satisfies at least one of the following conditions: the one-dimensional carbon material includes at least one of carbon nanotubes or carbon fibers; the zero-dimensional carbon material includes at least one of acetylene black, furnace black, Ketjen black, or carbon quantum dots; the two-dimensional carbon material includes at least one of graphene and multilayer graphite sheet; or the three-dimensional carbon material includes at least one of mesophase carbon microbeads, natural graphite, artificial graphite, hard carbon, or porous activated carbon.

[0025] In some embodiments of the present disclosure, the polymer binder includes at least one of polyvinylidene fluoride, polytetrafluoroethylene, styrene-butadiene rubber, polyacrylates, polyacrylonitriles, polyacrylic acids, sodium polyacrylates, carboxymethyl cellulose, sodium alginate, gum arabic, xanthan gum, or guar gum.

[0026] In some embodiments of the present disclosure, the holes are formed in the surface of the carbon coating using a laser radiation.

[0027] In a second aspect of the present disclosure, the present disclosure provides a sodium-ion battery. According to an embodiment of the present disclosure, the sodium-ion battery includes the negative electrode plate described above. As a result, the cycling performance of the sodium-ion battery can be effectively improved.

[0028] In a third aspect of the present disclosure, the present disclosure provides an electric device. According to an embodiment of the present disclosure, the electric device includes the sodium-ion battery described above. As a result, the electric device possesses all the advantages of the sodium-ion battery, which are not repeated herein.

[0029] According to an embodiment of the present disclosure, referring to FIG. 1, the negative electrode plate includes a negative electrode current collector 1, and a carbon coating 2 disposed on at least a portion of a surface of the negative electrode current collector 1. The carbon coating 2 includes a carbon material, a solid electrolyte, and a polymer binder. A surface of the carbon coating 2 facing away from the negative electrode current collector 1 has holes 3 extending along a thickness direction (i.e., direction Y) of the carbon coating 2. A depth of the holes 3 is smaller than

a thickness of the carbon coating 2. Therefore, in the present disclosure, by providing the holes in the surface of the negative electrode plate, a contact angle between an electrolyte solution and a negative electrode plate surface can be effectively reduced, which enables the electrolyte solution to rapidly wet the surface of the negative electrode plate, thereby significantly enhancing the liquid absorption and retention capability of the negative electrode plate and reducing the ionic transport impedance on the negative electrode side. Meanwhile, by providing the holes in the surface of the negative electrode plate, a surface roughness of the negative electrode plate is increased, enabling a sodium metal layer deposited on a surface of the negative electrode plate to have a stronger bonding force to the surface of the negative electrode plate. In this way, issues such as peeling and pulverization of the sodium deposition layer during cycling can be prevented, and the irreversible capacity loss can be avoided. Furthermore, an electric field on the negative electrode plate surface can be adjusted, and the uniformity and compactness of the deposited sodium on the negative electrode plate surface can be improved. Additionally, the solid electrolyte in the carbon coating can further enhance an affinity between the carbon coating and the electrolyte solution, thereby improving a wettability and ionic migration rate on the negative electrode side.

[0030] It may be noted that in FIG. 1, the direction Y indicates the thickness direction of the negative electrode plate, and the direction X indicates a width direction of the negative electrode plate.

[0031] The principle by which the negative electrode plate provided in the present disclosure can realize the above beneficial effect is described in detail below.

[0032] In the related art, a negative electrode side of a sodium-ion battery without a negative active material is generally a metal foil or a modified foil, and thus the negative electrode side has poor liquid absorption and retention capability. At the same time, in order to stabilize the negative electrode interface, the cell is usually subjected to an electrical performance test under a great external pressure. During the cycling process, due to the weak liquid absorption capability of the negative electrode side per se, in combination with the extruding effect of the cycling volume expansion on the electrolyte solution, the amount of the electrolyte solution on the negative electrode side may become lower with the cycling, which in turn leads to the increasingly significant polarization of the cell and the acceleration of the cyclic degradation of the cell.

[0033] Additionally, in the related art, due to the overly smooth surface of the negative electrode plate, the sodium metal layer deposited on the surface during cycling tends to peel off and pulverize, which results in electric insulation caused by the loss of contact with the conductive network, thereby affecting the capacity and cycling life of the cell.

[0034] To solve the above technical problem, in the present disclosure, the carbon coating is provided on at least a portion of the surface of the negative electrode current collector, and the holes (i.e., straight holes) extending along the thickness direction of the carbon coating are provided in the surface of the carbon coating. The depth of the holes is smaller than the thickness of the carbon coating (i.e., the holes do not penetrate through the carbon coating). In this way, the contact angle between the electrolyte solution and the surface of the negative electrode plate can be effectively reduced, which enables the contact angle θ between the surface of the negative electrode plate and the electrolyte solution to satisfy $0^\circ \leq \theta \leq 10^\circ$ (or even $0^\circ \leq \theta \leq 5^\circ$), thereby ensuring rapid wetting of the electrolyte solution on the surface of the negative electrode plate. Thus, the liquid absorption and retention capability of the negative electrode plate is significantly enhanced, and the ionic transport impedance on the negative electrode side is reduced. The negative electrode plate, when used in a sodium-ion battery, can effectively improve the cycling performance of the sodium-ion battery. Meanwhile, by providing the holes in the surface of the negative electrode plate, a surface roughness of the negative electrode plate is increased to be in the range of $0.4 \mu\text{m} \leq \text{Ra} \leq 3.2 \mu\text{m}$ (or even $2.4 \mu\text{m} \leq \text{Ra} \leq 3.2 \mu\text{m}$), enabling a sodium metal layer deposited on a surface of the negative electrode plate to have a stronger bonding force to the surface of the negative electrode plate. In this way, issues such as peeling and pulverization of the sodium deposition layer during

cycling can be prevented, and the irreversible capacity loss can be avoided. In addition, by providing the holes in the surface of the negative electrode plate, the electric field on the negative electrode plate surface can be adjusted, and the uniformity and compactness of the deposited sodium on the negative electrode plate surface can be improved.

[0035] Additionally, the carbon coating contains the carbon material, the solid electrolyte, and the polymer binder. Through the design of the carbon coating, the electrical conductivity of the negative electrode side is enhanced. Meanwhile, the solid electrolyte in the carbon coating can further enhance an affinity between the carbon coating and the electrolyte solution, thereby improving a wettability and ionic migration rate on the negative electrode side.

[0036] According to some specific embodiments of the present disclosure, the depth of the holes is denoted as $h_{\text{sub.2}}$, and the thickness of the carbon coating is denoted as $h_{\text{sub.1}}$, where $h_{\text{sub.2}}$ and $h_{\text{sub.1}}$ satisfy: $40\% \leq h_{\text{sub.2}}/h_{\text{sub.1}} \leq 80\%$. By limiting the depth of the holes within the above range, it can be further effectively ensured that the contact angle between the electrolyte solution and the surface of the negative electrode plate is reduced while the surface roughness of the negative electrode plate is increased. The inventors have found that if the depth of the holes is excessively small, i.e., insufficient drilling depth, the absorption amount of the electrolyte solution will be inadequate, thereby affecting the cycling performance of the sodium-ion battery. If the depth of the holes is excessively great, i.e., the holes are extremely close to the negative electrode current collector, foil leakage may occur, thereby affecting the metal deposition at the positions where the foil leakage occurs.

[0037] According to some further specific embodiments of the present disclosure, the holes may have a diameter ranging from $1\text{ }\mu\text{m}$ to $3\text{ }\mu\text{m}$, and a density of the holes in the surface of the carbon coating may range from 1,000 holes/ mm^2 to 3,000 holes/ mm^2 . By limiting the diameter and density of the holes within the above ranges, it can be further effectively ensured that the contact angle between the electrolyte solution and the surface of the negative electrode plate is reduced while the surface roughness of the negative electrode plate is increased.

[0038] In an embodiment of the present disclosure, the provision of the holes in the surface of the negative electrode plate and the addition of solid electrolyte to the carbon coating can both effectively reduce the contact angle between the electrolyte solution and the surface of the negative electrode plate, and in this case, the contact angle θ between the surface of the negative electrode plate and the electrolyte solution can be maintained in the range of $0^\circ \leq \theta \leq 10^\circ$ (or even $0^\circ \leq \theta \leq 5^\circ$), ensuring rapid wetting of the electrolyte solution on the surface of the negative electrode plate. In this way, the liquid absorption and retention capability of the negative electrode plate can be significantly enhanced, and the ionic transport impedance on the negative electrode side can be reduced.

[0039] It may be noted that the contact angle refers to an angle θ defined between a tangent to the gas-liquid interface at the liquid phase and a solid-liquid interface line, serving as a measure of wettability, where the tangent to the gas-liquid interface is taken at an intersection of gas, liquid, and solid phases.

[0040] In an embodiment of the present disclosure, by providing the holes in the surface of the roughness of the surface of the negative electrode plate can be effectively reduced, allowing the roughness of the surface of the negative electrode to be in the range of $0.4\text{ }\mu\text{m} \leq Ra \leq 3.2\text{ }\mu\text{m}$ (or even $2.4\text{ }\mu\text{m} \leq Ra \leq 3.2\text{ }\mu\text{m}$). Thus, the sodium metal layer deposited on the surface of the negative electrode plate has stronger binding force to the surface of the negative electrode plate, thereby effectively preventing issues such as peeling and pulverization of the sodium deposition layer during cycling and avoiding irreversible capacity loss.

[0041] In an embodiment of the present disclosure, the thickness $h_{\text{sub.1}}$ of the carbon coating is not greater than $5\text{ }\mu\text{m}$.

[0042] According to some further specific embodiments of the present disclosure, a weight ratio of the carbon material, the solid electrolyte, and the polymer binder may be (60 to 80):(0.5 to 10):(10

to 30). By limiting the weight ratio of the carbon material, the solid electrolyte, and the polymer binder within the above range, it can be further ensured that the electrical conductivity of the negative electrode side is enhanced and that the affinity between the carbon coating and the electrolyte solution is improved, thereby further enhancing the wettability and increasing the ion migration rate on the negative electrode side.

[0043] According to some further specific embodiments of the present disclosure, the carbon material is a combination of a one-dimensional carbon material and at least one selected from the group consisting of zero-dimensional carbon material, two-dimensional carbon material, and three-dimensional carbon material. The one-dimensional carbon material accounts for 60 wt % to 90 wt % based on the total weight of the carbon material as 100%. As a result, through the combination of carbon materials of different dimensions, the carbon coating can form a three-dimensional structure, which further enhances the liquid retention capability within the carbon coating and facilitates the deposition of sodium metal within the carbon coating, thereby reducing the deposition volume of sodium metal on the surface of the negative electrode plate and mitigates volume expansion.

[0044] The one-dimensional carbon material includes, but is not limited to, at least one of carbon nanotubes or carbon fibers, preferably carbon fibers, for providing a framework throughout the coating. The zero-dimensional carbon material includes, but is not limited to, at least one of acetylene black, furnace black, Ketjen black, and carbon quantum dots. The two-dimensional carbon material includes, but is not limited to, at least one of graphene or multilayer graphite sheet. The three-dimensional carbon material includes, but is not limited to, at least one of mesophase carbon microbeads, natural graphite, artificial graphite, hard carbon, or porous activated carbon. The combination of carbon fibers and acetylene black is preferred.

[0045] In an embodiment of the present disclosure, the specific type of the solid electrolyte is not particularly limited. Those skilled in the art may select as needed. Solid oxide electrolytes may be selected, such as β'' -Al₂O₃, NZSP-type, NLZSP-type (e.g., Na₃Zr₂Si₂PO₁₂, Na_{3+ x} La _{x} Zr_{2 x} Si₂PO₁₂ (0< x ≤0.5) and the like). Alternatively, solid sulfide electrolytes may be selected, such as Na₃PS₄, Na₁₁Sn₂PS₄, Na₇P₃S₁₁ and the like. Alternatively, solid halide electrolytes may be selected, such as Na₃MX₆ and the like. Alternatively, solid anti-perovskite electrolytes may be selected, such as X₃BA and the like. Alternatively, solid borohydride electrolytes may be selected, such as Na₂(B _{n} H _{n}) and the like. Preferably, at least one of Na₃Zr₂Si₂PO₁₂ and Na_{3+ x} La _{x} Zr_{2 x} Si₂PO₁₂ (0< x ≤0.5) is selected.

[0046] In an embodiment of the present disclosure, the specific type of polymer binder is not particularly limited and those skilled in the art may select as needed. Examples of the polymer binder includes at least one of polyvinylidene fluoride (PVDF), polytetrafluoroethylene (PTFE), styrene-butadiene rubber, polyacrylates, polyacrylonitriles, polyacrylic acids, sodium polyacrylates, carboxymethyl cellulose, sodium alginate, arabic gum, xanthan gum, and guar gum.

[0047] According to some further specific embodiments of the present disclosure, the holes may be formed in the surface of the carbon coating using a laser radiation, thereby further effectively ensuring the reduction of the contact angle between the electrolyte solution and the surface of the negative electrode plate and the increasing of the roughness of the surface of the negative electrode plate.

[0048] The preparation method for the aforementioned negative electrode plate includes: 1) mixing a carbon material, a solid electrolyte, and a polymer binder uniformly in a predetermined ratio, adding a solvent and stirring evenly to form a slurry, coating the slurry onto the negative electrode current collector, drying for subsequent use; and 2) forming the holes (i.e., straight holes) in a surface of the carbon coating using a laser radiation to obtain the negative electrode plate of the present disclosure.

[0049] The present disclosure further provides a sodium-ion battery. According to an embodiment of the present disclosure, the sodium-ion battery includes the negative electrode plate described above. As a result, the cycling performance of the sodium-ion battery can be effectively improved. [0050] Specifically, the sodium-ion battery described above may be a sodium-ion battery without a negative active material, which refers to a battery in which sodium ions are de-intercalated and in situ deposited onto the current collector of the negative electrode (in the present disclosure, sodium ions are deposited onto the carbon coating on the surface of the negative electrode current collector). During production and manufacturing, no negative active material is added, and the negative electrode current collector is used solely as the nominal negative electrode. However, such a negative electrode current collector does not have the function of a negative electrode. Only when metal migrates from the positive electrode material to the surface of the negative electrode current collector subsequent to the charging (in the present disclosure, migrating to the carbon coating on the surface of the negative electrode current collector), a metal layer can be formed on the negative electrode current collector to enable the negative electrode current collector to become a true negative electrode.

[0051] Specifically, the sodium-ion battery without a negative active material includes a positive electrode plate, a separator, a negative electrode plate, and an electrolyte solution.

[0052] The separator is disposed between the positive electrode plate and the negative electrode plate. The material of the separator is not specifically limited. As specific examples, the separator includes at least one of PP separator, PE separator, single-sided ceramic separator, double-sided ceramic separator, non-woven fabric separator, or glass fiber separator.

[0053] The positive electrode plate includes a positive electrode current collector and a positive electrode material layer formed on the positive electrode current collector, and the positive electrode material layer includes a positive active material, a positive electrode binder, and a positive conductive agent.

[0054] For the sodium-ion battery, the positive active material may include, but not limited to, one or more of transition metal oxides, polyanionic compounds, organic polymers, or Prussian blue-types materials. For the sodium transition metal oxides, the transition metal may include at least one of Mn, Fe, Ni, Co, Cr, Cu, Ti, Zn, V, Zr, or Ce. Examples of the sodium transition metal oxides include $\text{Na}_{x}\text{MO}_{2}$, where M is one or more of Ti, V, Mn, Co, Ni, Fe, Cr, and Cu, with $0 < x \leq 1$, such as $\text{Na}[\text{Ni}_{1/3}\text{Fe}_{1/3}\text{Mn}_{1/3}]\text{O}_{2}$.

[0055] The polyanionic compounds mentioned above are compounds with sodium ion, transition metal ion, and tetrahedral $(\text{YO}_{4})^{n-}$ anion unit. The transition metal may be at least one of Mn, Fe, Ni, Co, Cr, Cu, Ti, Zn, V, Zr, or Ce. Y may be at least one of P, S, or Si, and n represents the valence of $(\text{YO}_{4})^{n-}$, such as $\text{Na}_{3}\text{V}_{2}(\text{PO}_{4})_{3}$.

[0056] The Prussian blue-types compounds mentioned above are compounds containing sodium ion, transition metal ion, and cyanide ion (CN), the chemical formula of which can be represented by $\text{Na}_{x}\text{M1}[\text{M2}(\text{CN})_{6}]$, where $0 < x \leq 2$, and M1 and M2 are each at least one of Ni, Cu, Fe, Mn, Co, or Zn.

[0057] The specific material of the above positive electrode binder is not particularly limited. As specific examples, the positive electrode binder may include, but not limited to, one or more of polyvinylidene fluoride (PVDF), polytetrafluoroethylene (PTFE), polyacrylonitrile (PAN), and polyacrylic acid (PAA).

[0058] The specific material of the positive conductive agent is not particularly limited. As specific examples, the positive conductive agent may include, but not limited to, one or more of acetylene black, conductive carbon black, carbon nanotubes, carbon fibers, and graphene.

[0059] In an embodiment of the present disclosure, the specific material of the positive electrode current collector is not particularly limited. As specific examples, the positive electrode current collector may include, but not limited to, at least one of aluminum foil, carbon-coated aluminum foil, or stainless steel foil.

[0060] In an embodiment of the present disclosure, the above electrolyte solution includes an electrolyte salt and an organic solvent, where the specific types and compositions of the electrolyte salt and organic solvent are conventional choices in the field of batteries and can be selected according to actual needs.

[0061] In some embodiments, the electrolyte salt may include, but not limited to, at least one of sodium hexafluorophosphate (NaPF₆), sodium tetrafluoroborate (NaBF₄), sodium perchlorate (NaClO₄), sodium bis(trifluoromethanesulfonyl) imide (NaN(CF₃SO₂)₂), sodium bis(fluorosulfonyl) imide (NaN(SO₂F)₂), sodium borate bis(oxalate) (NaB(C₂O₄)₂), or sodium borate difluorooxalate (NaBF₂C₂O₄).

[0062] In some embodiments, the organic solvent may include, but not limited to, at least one of ethylene carbonate (EC), propylene carbonate (PC), diethyl carbonate (DEC), ethyl methyl carbonate (EMC), dimethyl carbonate (DMC), fluoroethylene carbonate (FEC), diethylene glycol dimethyl ether (DEGDME), ethylene glycol dimethyl ether (DME), 1,3-dioxolane (DOL), or tetraethylene glycol dimethyl ether (TEGDME).

[0063] The sodium-ion battery in the present disclosure may include the forms of a battery cell, a battery module, and a battery pack. In some embodiments, the battery cell can be assembled into a battery module. The number of battery cells included in the battery module can be one or more, and the specific number can be selected by those skilled in the art based on the application and capacity of the battery module. In some embodiments, the battery modules can further be assembled into a battery pack. The number of battery modules included in the battery pack can be one or more, and the specific number can be selected by those skilled in the art based on the application and capacity of the battery pack.

[0064] The present disclosure further provides an electric device. According to an embodiment of the present disclosure, the electric device includes the sodium-ion battery described above. As a result, the electric device possesses all the advantages of the sodium-ion battery, which will not be repeated here.

[0065] Specifically, the electric device mentioned above can include, but is not limited to, mobile phone, tablet, laptop, electric toy, electric tool, electric bicycle, electric vehicle, ship, spacecraft, and the like. Among these, electric toy may include stationary or mobile electric toys, such as gaming console, electric car toy, electric boat toy, and electric airplane toy, and the like, and spacecraft can include airplane, rocket, space shuttle, and space capsule, and the like.

[0066] The following describes the embodiments of the present disclosure in detail. It may be noted that the embodiments described below are exemplary and are only intended to explain the present disclosure, not to limit it. Additionally, unless otherwise specified, all reagents used in the following examples are commercially available or can be synthesized according to methods described herein or known methods. For reaction conditions not listed, they are easily accessible to those skilled in the art.

Example 1

[0067] This Example provided a sodium-ion battery, the preparation method of which included the following steps:

(1) Preparation of Negative Electrode Plate

[0068] Carbon nanotubes, a polyacrylate binder, and β'' -Al₂O₃ were first dispersed into NMP solvent and mixed uniformly to obtain a carbon coating slurry, in which a weight ratio of carbon nanotubes, polyacrylate, and β'' -Al₂O₃ was 70:25:5. The carbon coating slurry was coated onto the aluminum foil negative current collector and subjected to drying, cold pressing, slitting, cutting into sheets for subsequent use. The carbon coating has a thickness of 5 μ m.

[0069] Then, straight holes with a diameter of 2 μ m and a depth of 3 μ m were formed in the surface of the carbon coating using a laser radiation with a density of 2,000 holes/mm², to obtain the

negative electrode plate of the present disclosure.

(2) Preparation of Positive Electrode Plate

[0070] A positive active material ($\text{Na}_4\text{Fe}_3(\text{PO}_4)_2\text{P}_2\text{O}_7$), a conductive agent (Super-P), and a binder (PVDF) were mixed in a weight ratio of 96:2:2. The mixed powder was placed in a vacuum mixer and added with N-methyl-2-pyrrolidone (NMP). The mixture was stirred uniformly to obtain a positive electrode slurry. The positive electrode slurry was evenly coated on both sides of an aluminum foil (i.e., a positive electrode current collector). The coated positive electrode current collector was transferred to an oven for drying, then subjected to rolling, and cutting to obtain the positive electrode plate

(3) Preparation of Separator

[0071] A polyethylene film with a thickness of 17 μm was used as the separator.

(4) Preparation of Electrolyte Solution

[0072] The dry sodium salt (NaPF_6) was added in a solvent mixture of diethylene glycol dimethyl ether and tetraethylene glycol dimethyl ether to prepare an electrolyte solution with a concentration of 1 mol/L. A volume ratio of diethylene glycol dimethyl ether to tetraethylene glycol dimethyl ether was 1:1.

(5) Preparation of Sodium-Ion Battery:

[0073] The prepared positive electrode plate, separator, and negative electrode plate were stacked in sequence, to allow the separator to be disposed between the positive and negative electrode plates, and wound to obtain a bare battery cell. The bare battery cell was placed into an aluminum-plastic pouch and subjected to drying, and then the electrolyte solution was injected, followed by processes of encapsulating in vacuum, standing, forming and shaping to obtain the pouch sodium-ion battery.

Example 2

[0074] Examples 2 merely differed from Example 1 in that straight holes with a diameter of 1 μm and a depth of 4 μm were formed in the surface of the carbon coating using a laser radiation with a density of 3,000 holes/ mm^2 . The rest part of Example 2 was the same as those of Example 1.

Example 3

[0075] Example 3 merely differed from Example 1 in that straight holes with a diameter of 3 μm and a depth of 2 μm were formed in the surface of the carbon coating using a laser radiation with a density of 1,000 holes/ mm^2 . The rest part of Example 3 was the same as those of Example 1.

Example 4

[0076] Examples 4 merely differed from Example 1 in that $\beta''\text{-Al}_2\text{O}_3$ was replaced with $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$. The rest part of Example 4 was the same as those of Example 1.

Example 5

[0077] Examples 5 merely differed from Example 1 in that $\beta''\text{-Al}_2\text{O}_3$ was replaced with $\text{Na}_3.25\text{La}_{0.25}\text{Zr}_{0.5}\text{Si}_2\text{PO}_{12}$. The rest part of Example 5 was the same as those of Example 1.

Example 6

[0078] Examples 6 merely differed from Example 1 in that a weight ratio of carbon nanotubes, polyacrylate and $\beta''\text{-Al}_2\text{O}_3$ was 70:28:2. The rest part of Example 6 was the same as those of Example 1.

Example 7

[0079] Examples 7 merely differed from Example 1 in that a weight ratio of carbon nanotubes, polyacrylate and $\beta''\text{-Al}_2\text{O}_3$ was 70:20:10. The rest part of Example 7 was the same as those of Example 1.

Example 8

[0080] Examples 8 merely differed from Example 1 in that the carbon nanotubes in the carbon coating slurry were replaced with a combination of carbon nanotubes and acetylene black at a

weight ratio of carbon nanotubes to acetylene black of 80:20. The rest part of Example 8 was the same as those of Example 1.

Example 9

[0081] Examples 9 merely differed from Example 1 in that replacing the carbon nanotubes in the carbon coating slurry in this Example with a combination of carbon nanotubes and graphene at a weight ratio of carbon nanotubes and graphene of 70:30. The rest part of Example 9 was the same as those of Example 1.

Example 10

[0082] Examples 10 merely differed from Example 1 in that the carbon nanotubes in the carbon coating slurry were replaced with a combination of carbon nanotubes and mesophase carbon microbeads at a weight ratio of carbon nanotubes to mesophase carbon microbeads of 60:40. The rest part of Example 10 was the same as those of Example 1.

Example 11

[0083] Examples 11 merely differed from Example 4 in that the carbon nanotubes in the carbon coating slurry were replaced with a combination of carbon fibers and acetylene black at a weight ratio of carbon fibers to acetylene black of 80:20. The rest part of Example 11 was the same as those of Example 4.

Comparative Example 1

[0084] Comparative Example 1 merely differed from Example 1 in (1) preparation of the negative electrode plate: not forming straight holes in the surface of the carbon coating. The rest part of Comparative Example 1 was the same as those of Example 1.

Comparative Example 2

[0085] Comparative Example 2 merely differed from Example 1 in (1) preparation of the negative electrode plate: not adding β'' -Al₂O₃ to the carbon coating slurry. The rest part of Comparative Example 2 was the same as those of Example 1.

[0086] The contact angle between the electrolyte solution and the surface of the negative electrode plates prepared in Examples 1 to 11 and Comparative Examples 1 to 2 was measured according the following method: 1) placing a sample of negative electrode plate on a sample stage of a contact angle measurement device and adjusting the sample appropriately to ensure that the surface of the sample of negative electrode plate was at the same level as the liquid in the syringe; 2) slowly injecting the electrolyte solution onto the surface of the negative electrode plate sample using a syringe while ensuring that the liquid did not flow or spread on the surface; 3) capturing images of the liquid droplet on the sample surface including front and side views of the liquid droplet using the device's camera to obtain complete information of the contact angle; and 4) measuring the contact angle using the measurement software's ruler tool, and performing multiple measurements to calculate the average to improve the accuracy. The measurement results were shown in Table 1.

[0087] The surface roughness of the negative electrode plates in Examples 1 to 11 and Comparative Examples 1 to 2 was measured according the following method: taking the cut negative electrode plate and calculating the surface roughness of the negative electrode plate using the ISO 25178 standard conforming to international specification for the measurement of microscopic surface geometry to obtain the parameter Ra. The measurement results were shown in Table 1.

[0088] The binding force between the sodium metal layer deposited on surface of the negative electrode plate and the surface of the negative electrode plate in Examples 1 to 11 and Comparative Examples 1 to 2 was measured according the following method: purchasing tapes with different adhesion strengths (furrow tape, chloroprene adhesive, and blue glue, and for the adhesion strength of the tapes, blue glue > chloroprene adhesive > furrow tape), disassembling fully charged cells in a glovebox, adhering tapes with different adhesion strengths to the surface of the deposition layer on the negative electrode, rolling back and forth three times with a 2 kg roller, performing peel test using a small tensile strength machine, determining the condition of the sodium metal layer adhered to the tape to judge the adhesion. The measurement results were shown in Table 1.

[0089] The cycling performance of the sodium-ion batteries prepared in Examples 1-11 and Comparative Examples 1-2 was tested according the following method: applying a 0.25 MPa pre-tightening force to the cells using flat testing fixture for cycling test at a charge-discharge rate of 0.5P/0.5P within a voltage range controlled between 2.5V and 3.5V at room temperature, and recording the number of cycles at 90% loss of cycling capacity. The measurement results were shown in Table 1.

TABLE-US-00001

Number	Example	Contact angle/°	Roughness/ μm	Adhesion of cycles	Description
1	4.2	2.8	Small amount of sodium layer was 453 found on chloroprene adhesive, and sodium layer was found on a great area of blue glue		
2	4.6	2.6	Small amount of sodium layer was 417 found on chloroprene adhesive, and sodium layer was found on a great area of blue glue		
3	4.3	2.9	Small amount of sodium layer was 423 found on chloroprene adhesive, and sodium layer was found on a great area of blue glue		
4	3.6	2.8	No significant sodium layer 512	adhesion was found on blue glue	
5	3.9	2.9	No significant sodium layer 508	adhesion was found on blue glue	
6	4.8	2.7	Small amount of sodium layer was 383 found on chloroprene adhesive, and sodium layer was adhered to the whole area of blue glue		
7	4.1	2.6	No significant sodium layer 467	adhesion was found on chloroprene adhesive, and small amount of sodium layer was found on blue glue	
8	4.0	2.4	Small amount of sodium layer was 468 found on chloroprene adhesive, and small amount of sodium layer was found on blue glue		
9	4.1	2.6	Small amount of sodium layer was 459 found on chloroprene adhesive, and sodium layer was adhered to the whole area of blue glue		
10	4.1	2.7	Small amount of sodium layer was 455 found on chloroprene adhesive, and sodium layer was adhered to the whole area of blue glue		
11	3.4	2.6	No significant sodium layer 593	adhesion was found on blue glue	
Comparative Example 1	17	2.0	Significant sodium layer adhesion 264		
Comparative Example 2	8.7	2.7	Small amount of scattered sodium 324	layer was found on furrow tape, and small amount of sodium layer was found on chloroprene adhesive	

[0090] Table 1 reveals that, compared to Comparative Example 1, each of the negative electrode plates of Examples 1 to 11 exhibited significantly reduced contact angle between the negative electrode plate and the electrolyte solution, significantly increased roughness of the surface of the negative electrode plate, significantly improved adhesion between the sodium metal layer deposited on the surface of the negative electrode plate and the negative electrode plate, and significantly enhanced cycling performance of the battery. These results indicate that the provision of holes in the surface of the negative electrode plate can significantly reduce the contact angle between the negative electrode plate and the electrolyte solution, significantly increase surface roughness, significantly improve the adhesion between the sodium metal layer deposited on the surface of the negative electrode plate and the negative electrode plate, and significantly enhance the cycling performance of the battery.

[0091] Table 1 further reveals that, compared to Comparative Example 2, each of the negative electrode plates of Examples 1 to 11 exhibited significantly reduced contact angles between the negative electrode plate and the electrolyte solution, and significantly enhanced cycling performance of the battery. These results indicate that the addition of the solid electrolyte to the carbon coating can effectively reduce the contact angle between the negative electrode plate and the electrolyte solution and enhance the cycling performance of the battery.

[0092] Furthermore, Table 1 reveals that, compared to Example 1, each of the negative electrode plates of Examples 8 to 10 exhibited reduced contact angles between the negative electrode plate and the electrolyte solution, and improved cycling performance of the battery. These results indicate that the combination of as carbon nanotubes as the one-dimensional carbon material and acetylene black the zero-dimensional carbon material, the combination of carbon nanotubes as the one-dimensional carbon material and graphene as the two-dimensional carbon material, or the combination of carbon nanotubes as the one-dimensional carbon material and mesophase carbon

microbeads as the three-dimensional carbon material can enhance the affinity between the carbon coating and the electrolyte solution and improve the cycling performance of the battery.

[0093] Table 1 further reveal that, compared to Example 4, the negative electrode plate of Example 11 exhibited a significantly reduced contact angle between the negative electrode plate and the electrolyte solution, and significantly improved cycling performance of the battery. These results indicate that the combination of carbon fibers as the one-dimensional carbon materials and acetylene black as the zero-dimensional carbon material can significantly enhance the affinity between the carbon coating and the electrolyte solution, and improve the cycling performance of the battery. Moreover, Example 11 demonstrates the best overall performance, indicating that the optimal overall performance of the battery can be achieved by using $\text{Na.sub.3Zr.sub.2Si.sub.2PO.sub.12}$ as the solid electrolyte and the combination of carbon fibers as the one-dimensional carbon materials and acetylene black as the zero-dimensional carbon material.

[0094] Throughout the specification, the term such as “one embodiment” “some embodiments” “example” “specific example” or “some examples” is intended to indicate that the specific features, structures, materials, or characteristics described in connection with the embodiment or example are included in at least one embodiment or example of the present disclosure. In the specification, the indicative descriptions of the above terms do not necessarily refer to the same embodiment or example. Furthermore, the specific features, structures, materials, or characteristics described can be combined in any suitable manner in one or more embodiments or examples. Additionally, unless there are contradictions, those skilled in the art can combine and integrate the different embodiments or examples and the features of different embodiments or examples described in this specification.

[0095] Although the embodiments of the present disclosure have been illustrated and described above, it may be understood that these embodiments are exemplary and may not be construed as limiting the present disclosure. Those skilled in the art can make variations, modifications, substitutions, and changes to the embodiments described above within the scope of the present disclosure.

Claims

1. A negative electrode plate, comprising: a negative electrode current collector; and a carbon coating, disposed on at least a portion of a surface of the negative electrode current collector, wherein: the carbon coating comprises a carbon material, a solid electrolyte, and a polymer binder; a surface of the carbon coating facing away from the negative electrode current collector has holes extending along a thickness direction of the carbon coating, a depth of the holes being smaller than a thickness of the carbon coating.
2. The negative electrode plate according to claim 1, wherein: the depth of the holes is denoted as $h_{\text{sub.2}}$, and the thickness of the carbon coating is denoted as $h_{\text{sub.1}}$, $h_{\text{sub.2}}$ and $h_{\text{sub.1}}$ satisfying: $40\% \leq h_{\text{sub.2}}/h_{\text{sub.1}} \leq 80\%$.
3. The negative electrode plate according to claim 2, wherein the holes satisfy at least one of the following conditions: the holes each have a diameter ranging from 1 μm to 3 μm ; or a density of the holes on the surface of the carbon coating ranges from 1,000 holes/ mm^2 to 3,000 holes/ mm^2 .
4. The negative electrode plate according to claim 1, wherein a contact angle θ between a surface of the negative electrode plate and an electrolyte solution satisfies: $0^\circ \leq \theta \leq 10^\circ$.
5. The negative electrode plate according to claim 1, wherein a roughness R_a of a surface of the negative electrode plate satisfies: $0.4 \mu\text{m} \leq R_a \leq 3.2 \mu\text{m}$.
6. The negative electrode plate according to claim 2, wherein the thickness $h_{\text{sub.1}}$ of the carbon coating is not greater than 5 μm .
7. The negative electrode plate according to claim 1, wherein a weight ratio of the carbon material,

the solid electrolyte, and the polymer binder is (60 to 80):(0.5 to 10):(10 to 30).

8. The negative electrode plate according to claim 1, wherein the carbon material is a combination of a one-dimensional carbon material and at least one selected from zero-dimensional carbon material, two-dimensional carbon material, and three-dimensional carbon material.

9. The negative electrode plate according to claim 8, wherein the carbon material satisfies at least one of the following conditions: the one-dimensional carbon material comprises at least one of carbon nanotubes or carbon fibers; the zero-dimensional carbon material comprises at least one of acetylene black, furnace black, Ketjen black, or carbon quantum dots; the two-dimensional carbon material comprises at least one of graphene or multilayer graphite sheet; or the three-dimensional carbon material comprises at least one of mesophase carbon microbeads, natural graphite, artificial graphite, hard carbon, or porous activated carbon.

10. The negative electrode plate according to claim 1, wherein the solid electrolyte comprises at least one of β'' -Al₂O₃, Na₃Zr₂Si₂PO₁₂, Na_{3+x}La_xZr₂Si₂PO₁₂, Na₃PS₄, Na₁₁Sn₂PS₄, or Na₇P₃S₁₁, where $0 < x \leq 0.5$.

11. The negative electrode plate according to claim 1, wherein the polymer binder comprises at least one of polyvinylidene fluoride, polytetrafluoroethylene, styrene-butadiene rubber, polyacrylates, polyacrylonitriles, polyacrylic acids, sodium polyacrylates, carboxymethyl cellulose, sodium alginate, gum arabic, xanthan gum, or guar gum.

12. The negative electrode plate according to claim 1, wherein the holes are formed in the surface of the carbon coating using laser radiation.

13. A sodium-ion battery, comprising: a positive electrode plate; a separator; a negative electrode plate; and an electrolyte solution, wherein the negative electrode plate comprises a negative electrode current collector and a carbon coating disposed on at least a portion of a surface of the negative electrode current collector, wherein: the carbon coating comprises a carbon material, a solid electrolyte, and a polymer binder; a surface of the carbon coating facing away from the negative electrode current collector has holes extending along a thickness direction of the carbon coating, a depth of the holes being smaller than a thickness of the carbon coating.

14. The sodium-ion battery according to claim 13, wherein: the depth of the holes is denoted as h_2 , and the thickness of the carbon coating is denoted as h_1 , h_2 and h_1 satisfying: $40\% \leq h_2/h_1 \leq 80\%$.

15. The sodium-ion battery according to claim 14, wherein the holes satisfy at least one of the following conditions: the holes have each a diameter ranging from 1 μm to 3 μm ; and a density of the holes on the surface of the carbon coating ranges from 1,000 holes/ mm^2 to 3,000 holes/ mm^2 .

16. The sodium-ion battery according to claim 13, wherein a contact angle θ between a surface of the negative electrode plate and an electrolyte solution satisfies: $0^\circ \leq \theta \leq 10^\circ$.

17. The sodium-ion battery according to claim 13, wherein a roughness R_a of a surface of the negative electrode plate satisfies: $0.4 \mu\text{m} \leq R_a \leq 3.2 \mu\text{m}$.

18. The sodium-ion battery according to claim 14, wherein the thickness h_1 of the carbon coating is not greater than 5 μm .

19. The sodium-ion battery according to claim 13, wherein a weight ratio of the carbon material, the solid electrolyte, and the polymer binder is (60 to 80):(0.5 to 10):(10 to 30).

20. An electric device, comprising the sodium-ion battery according to claim 13.
